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Mechanistic Middle-Out Physiologically Based Toxicokinetic Modeling of Transporter-Dependent Disposition of Perfluorooctanoic Acid in Humans

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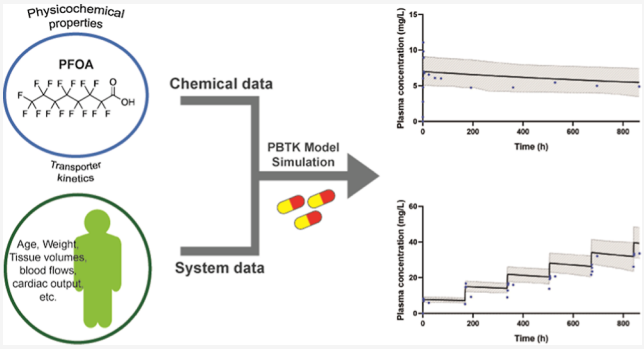
ABSTRACT: Perfluorooctanoic acid (PFOA) is an environmental toxicant exhibiting a years-long biological half-life ($t_{1/2}$) in humans and is linked with adverse health effects. However, limited understanding of its toxicokinetics (TK) has obstructed the necessary risk assessment. Here, we constructed the first middle-out physiologically based toxicokinetic (PBTK) model to mechanistically explain the persistence of PFOA in humans. In vitro transporter kinetics were thoroughly characterized and scaled up to in vivo clearances using quantitative proteomics-based in vitro-to-in vivo extrapolation. These data and physicochemical parameters of PFOA were used to parameterize our model. We uncovered a novel uptake transporter for PFOA, highly likely to be monocarboxylate transporter 1 which is ubiquitously expressed in body tissues and may mediate broad tissue penetration. Our model was able to recapitulate clinical data from a phase I dose-escalation trial and divergent half-lives from clinical trial and biomonitoring studies. Simulations and sensitivity analyses confirmed the importance of renal transporters in driving extensive PFOA reabsorption, reducing its clearance and augmenting its $t_{1/2}$. Crucially, the inclusion of a hypothetical, saturable renal basolateral efflux transporter provided the first unified explanation for the divergent $t_{1/2}$ of PFOA reported in clinical (116 days) versus biomonitoring studies (1.3−3.9 years). Efforts are underway to build PBTK models for other perfluoroalkyl substances using similar workflows to assess their TK profiles and facilitate risk assessments.



KEYWORDS: *PFOA, PBTK model, in vitro-to-in vivo extrapolation, quantitative proteomics, renal transporters, half-life ($t_{1/2}$)*

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1. INTRODUCTION

Perfluorooctanoic acid (PFOA) is a prominent environmental toxicant belonging to a group of synthetic compounds known as per- and polyfluoroalkyl substances (PFAS). Used extensively in many industrial and consumer products since the 1950s,[1] humans are exposed to PFOA through these products, as well as contaminated water and food sources.[2,3] The ban on PFOA production has yet to effect a global phase-out, and PFOA exposure continues to be pervasive with its widespread pollution. In humans, PFOA has been detected not just in 98% of blood samples tested in the US population[4] but also in breast milk[5] and umbilical cord blood.[6]

With an extremely long half-life ($t_{1/2}$) of 1.3−3.9 years in humans,[7−13] the bioaccumulation of PFOA is concerning as chronic exposure to PFOA has been associated with cancer, thyroid hormone disruption, immunological suppression, reduced kidney function, and reduced infant birth weights;[14−16] similar effects have been observed in animal studies.[14] Extensive C−F bonds render PFOA metabolically inert and highly resistant to degradation. As an acid with a low pK_a, PFOA is almost fully ionized at physiological pH[17] and is highly plasma protein bound,[18] resulting in poor cellular passive permeability. As such, the disposition of PFOA is heavily influenced by membrane transporters. For example, hepatic uptake of PFOA is facilitated by transporters such as organic anion transporting polypeptides (OATP) 1B1, 1B3, 2B1, and sodium taurocholate cotransporting polypeptide (NTCP) (Figure S1).[19,20] In the kidney, the organic anion transporter 1 (OAT1) and OAT3 facilitate active secretion, while OAT4 and urate transporter 1 (URAT1) drive tubular reabsorption (Figure S1).[21] Renal clearance through filtration and transporter-mediated secretion has been shown to drive PFOA elimination in animals;[22] conversely, greater tubular

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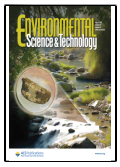

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active reabsorption is believed to decrease the renal elimination of PFOA in humans, resulting in a longer $t_{1/2}$.[21] While PFOA transport by these transporters has been measured individually, their aggregate influence on PFOA's disposition in humans and, in particular, its $t_{1/2}$ has not been examined quantitatively and mechanistically. Transporters also determine the extent of PFOA distribution into tissues, thereby influencing the volume of distribution (V_d) and likelihood of tissue effects.

Estimates for PFOA $t_{1/2}$ in animals range from hours and days in rats, weeks in mice, to months in monkeys, and up to several years in humans.[21] The poor inter-species correlation makes it difficult to extrapolate from animals to humans. Even across human studies, there is a vast difference in the estimated $t_{1/2}$.[23] Elcombe et al. performed a phase I dose-escalation trial on cancer patients, where the $t_{1/2}$ of PFOA was estimated as 116 days based on the plasma concentration−time measurements following PFOA exposure.[24] On the other hand, several biomonitoring studies reported years-long $t_{1/2}$ for PFOA. Li et al. calculated a mean $t_{1/2}$ of 2.7 years,[13] while Zhang et al. estimated a PFOA $t_{1/2}$ of 1.3 years from a one-compartmental model based on human blood and urine samples.[10]

Prospective toxicokinetic (TK) studies of PFOA are understandably limited due to ethical and safety concerns. Consequently, physiologically based toxicokinetic (PBTK) modeling has emerged as an alternative to predict TK profiles[25−27] and relate internal levels to external exposures for risk assessment.[28] PBTK modeling quantitatively simulates the absorption, distribution, metabolism, and excretion of compounds by creating a virtual, anatomically accurate human model that incorporates physiological and chemical processes influencing TK behaviour.[29] Previous PFOA PBTK modeling efforts either utilized components derived from animal data (e.g., kinetic parameters, renal transporter expression, or physicochemical properties)[25] or adopted a top-down approach via empirical fitting to human data.[10,27] However, the disconnect in transporter kinetics between animals and humans[21] reduces the reliability of such extrapolations, while the top-down approach is limited to the population and exposure scenario it is derived from and lacks mechanistic insight.

Here, we utilized middle-out PBTK modeling to predict the human TK of PFOA. The model is extensively parameterized by PFOA-specific physicochemical properties and in vitro human data, which are then scaled to in vivo levels using quantitative proteomics-based in vitro-to-in vivo extrapolation (IVIVE). We have previously demonstrated the robustness of this approach.[30] Our PFOA PBTK model was then validated against available human clinical data and used to uncover the mechanisms influencing the disposition of PFOA and its prolonged $t_{1/2}$. For the first time, our model reconciled the varying PFOA $t_{1/2}$ values observed across human studies. This work also sets forth a rigorous approach using in vitro transport kinetic data to predict and assess the TK of other PFAS chemicals.

2. MATERIALS AND METHODS

2.1. Uptake Studies. The transport kinetics of OATP1B1, OATP1B3, and OATP2B1 stably expressed in Chinese Hamster Ovary (CHO) cells and NTCP, OAT1, OAT3, OAT4, and URAT1 stably expressed in Human Embryonic Kidney 293 (HEK293) cells were determined using in vitro cell-based assays (Figure S1). Further description of cell culture and cell seeding for uptake studies can be found in Methods S1 and S2, respectively. The appropriate incubation time (uptake duration) for each transporter used was determined in Method S3. Concentration-dependent uptake assays were performed by varying PFOA concentrations at the optimized incubation time. Uptake was initiated by adding 200 μL of transport solution containing the substrate and incubating the cells at 37 °C. At the end of the incubation time, the uptake buffer was aspirated, and the cells were washed twice with 200 μL of ice-cold Dulbecco's phosphate-buffered saline to terminate uptake, before alkaline lysis via the addition of 100 μL of 0.2 N sodium hydroxide containing internal standards for 10 min with shaking on ice. The lysate was neutralized with 100 μL of 0.2 N hydrochloric acid, and 100 μL of acetonitrile was added for protein precipitation. The samples were well mixed, and a 50 μL aliquot was used for quantification of protein concentration. After centrifugation at 3220*g* for 30 min, the supernatant was subsequently collected and analyzed using liquid chromatography−tandem mass spectrometry (LC−MS/MS) as described in Method S4 to measure the intracellular concentration of PFOA. Calibration standards ranging from 10 to 4000 nM PFOA were prepared using matched cell lysate in sodium hydroxide, hydrochloric acid, and acetonitrile mixture. Protein concentration was determined using the Pierce BCA Protein Assay Kit, and quantification was done using the 2104 EnVision Multilabel Plate Reader (PerkinElmer, Waltham, MA) at 562 nm.

2.2. Kinetic Analysis. The intracellular concentrations measured from LC−MS/MS were normalized with the incubation time and protein content to derive the uptake rates. PFOA uptake rates measured in CHO cells were fitted into the following modified Michaelis−Menten equation

$$V = \frac{J_{max} \times [S]}{K_m + [S]} + CL_{PD} \times [S] \quad (1)$$

where V is the uptake rate (pmol/min/mg protein), J_{max} is the maximal uptake rate (pmol/min/mg protein), [S] is the substrate concentration (μM), K_m is the Michaelis−Menten constant representing the substrate affinity for a transporter (μM), and CL_{PD} is passive diffusion clearance (μL/min/mg protein). Due to the observed presence of two PFOA transporters (background and transfected transporters) in HEK293 cell lines, PFOA uptake rates measured in these cell lines were fitted into the modified two-transporter Michaelis−Menten equation to segregate the kinetic parameters of the transporter of interest from that of the background transporter

$$V = \frac{J_{max,1} \times [S]}{K_{m,1} + [S]} + \frac{J_{max,2} \times [S]}{K_{m,2} + [S]} + CL_{PD} \times [S] \quad (2)$$

where $J_{max,1}$ and $K_{m,1}$ represent the kinetics of the background transporter, while $J_{max,2}$ and $K_{m,2}$ represent that of the transporter of interest. $J_{max,1}$, $K_{m,1}$, and CL_{PD} were obtained from nonlinear regression of the saturable uptake observed in HEK293 wild-type cells using eq 1. These parameters were then substituted into eq 2, while $J_{max,2}$ and $K_{m,2}$ were left as unknowns and fitted using nonlinear regression.

Curve fitting and kinetic parameters were obtained via nonlinear regression using Prism (version 8.0, GraphPad Software, San Diego, CA).

2.3. PBTK Model Development. The population-based Simcyp simulator version 20 (Certara, Simcyp Division, Sheffield, UK) was used to develop our PBTK model of

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PFOA. A schematic diagram of the model structure and organ compartments is illustrated in Figure S2. A detailed description on the model structure and its differential equations can be found in the referenced articles.[31−33] Compound-specific parameters are listed in Table S2, and the compound file is provided in the Supporting Information. Oral absorption was modeled using the intestinal Advanced Dissolution, Absorption, and Metabolism (ADAM) model.[34] The apical uptake transporter OATP2B1 was incorporated in the ADAM model to permit active intestinal uptake. A full PBTK model was used to describe the perfusion-limited distribution of PFOA in all tissues except the liver and kidney where permeability-limited liver (PerL)[35] and kidney (Mechanistic Kidney Model, Mech KiM) models[36,37] were used to incorporate transporter-mediated disposition. The tissue-to-plasma distribution equilibrium ratios (K_p) were predicted using the method by Rodgers and Rowland.[38−40] Two hypothetical transporters, namely a hepatic sinusoidal and a renal basolateral efflux transporter, were incorporated into our model to permit PFOA exit from the liver and kidney. This was necessary as the absence of these transporters would result in perpetual PFOA accumulation within these tissues with no means of exit.

2.4. PBTK Model Validation Against a High-Dose PFOA Exposure. Simulations were performed for a single oral 50 mg PFOA dose and repeated weekly dosing of 50 mg PFOA for a virtual population of cancer patients (full physiological parameters provided in "Simulation output for 50 mg single dose" in Supporting Information) to match the dosing scenario to a phase I dose-escalation trial by Elcombe et al.[24] The observed data for the single and repeated weekly oral dose represented the mean plasma concentration−time profiles for three patients and a single patient, respectively. Further details of the trial design are tabulated in Table S3. The 50 mg single dose is also referred to as high-dose exposure hereafter. Our model was validated against the PFOA time course data. Separately, a two-compartmental analysis using the PKPlus module in GastroPlus version 9.8 (Simulations Plus, Lancaster, CA) was performed on the cancer trial clinical data to obtain TK parameters for PFOA such as C_{max} and area under the curve (AUC), which were not reported by Elcombe et al. Model validation was performed using the 2-fold criterion comparing TK parameters obtained from our simulation versus the clinical trial.[41]

2.5. Predicting Steady-State Plasma Concentrations (C_{ss}) at Low Daily PFOA Exposures. Tittlemier et al.[42] determined the average dietary exposure to PFOA in Canada to be 70 ng/day according to the total diet study in 2004. This estimate was used to perform a 11 years-long daily PFOA dose simulation (maximum permitted simulation time for the Simcyp simulator) in an adult healthy population representative. Trial design details are provided in Table S3. The 70 ng daily dose is also referred to as low-dose exposure hereafter. The population representative is a single virtual individual created using average values for the physiological parameters of the population (full parameters provided in "Simulation output for male and female representative with 70 ng daily dose" in Supporting Information). Unlike the simulation of the study by Elcombe et al., the population representative was used to simulate the exposure conditions reported by Tittlemier et al. because hardware and software limitations did not permit simulation of a full virtual population for 11 years. The predicted C_{ss} was compared to the reported values in Canadian Health Measures Survey (CHMS) 2007−2009.[43] The predicted $t_{1/2}$ was compared to that from the biomonitoring studies (1.3−3.9 years).[7−13]

2.6. Reverse Dosimetry to Estimate the Daily Intake in a Healthy Adult Female. The European Food Safety Authority (EFSA) Opinion 2020 estimated using a German study that the serum PFOA level in a healthy 35 year-old adult female is 2 ng/mL following a 0.187 ng/kg body weight per day of PFOA exposure (see Figure 12 of the Scientific Opinion).[28] We performed reverse dosimetry using our model to determine the daily intake of PFOA that would result in this serum level in an adult female. The simulation was performed using a female population representative aged 24 years with a body weight of 68 kg for a duration of 11 years (simulated subject reaches 35 years of age at the conclusion of the simulation). Full physiological data is provided in "Simulation output for female representative with daily dose of 0.28 ng per kg body weight" in Supporting Information.

2.7. Statistical Analysis. Data were analyzed using Student's *t*-test for transporter functionality tests. Differences were considered to be statistically significant for $p < 0.05$.

Additional materials and methods are given in the Supporting Information.

3. RESULTS

3.1. Uptake of PFOA by Hepatic and Renal Transporters. Functionalities of the shortlisted transporters were confirmed with known probe substrates prior to performing PFOA uptake assays (Figure S3), except for URAT1 as we were unable to detect uric acid using LC−MS/MS. Transport of PFOA by all selected transporters was confirmed by comparing the difference in uptake between wild-type and transfected cells as shown in Figure S3. PFOA linear uptake duration by OATP1B1, OATP1B3, OATP2B1, and NTCP was determined to be 1 min, whereas that by OAT1, OAT3, OAT4, and URAT1 was 2.5 min (Figure S4). Concentration-dependent uptakes of PFOA by the shortlisted transporters are shown in Figures S5 and S6. The measured kinetic parameters corresponding to each transporter are tabulated in Table S4.

3.2. Background Transport of PFOA in HEK293 Cells. The background transport of PFOA in HEK293 cells was investigated as described in Method S5. We observed background transport activity in HEK293 wild-type cells as illustrated by the saturable uptake kinetics in Figure S6 instead of linear profiles lines as observed in CHO wild-type cells in Figure S5. Uptake experiments performed at 4 °C in wild-type cells (Figure S7A) demonstrate a similar curved profile, ruling out ATP-dependent transporters as the background transporter. We speculate that PFOA is a novel and previously unreported substrate of the monocarboxylate transporter 1 (MCT1), a transporter present endogenously in many cells, including HEK293.[44] This is supported by the structural similarity between PFOA and octanoic acid, a known substrate of MCT1.[45,46] The presence of MCT1 initially confounded our results as its uptake effects in wild-type cells could not be distinguished from that of the overexpressed transporter. As published kinetic data for PFOA uptake did not account for the effects of MCT1, fresh kinetic data for the full panel of PFOA transporters were generated. To isolate the uptake kinetics of PFOA by our transporter of interest from MCT1, we (1) optimized our experimental conditions to increase the delta of PFOA uptake difference between MCT1 and the transfected transporter and (2) modified the Michaelis−

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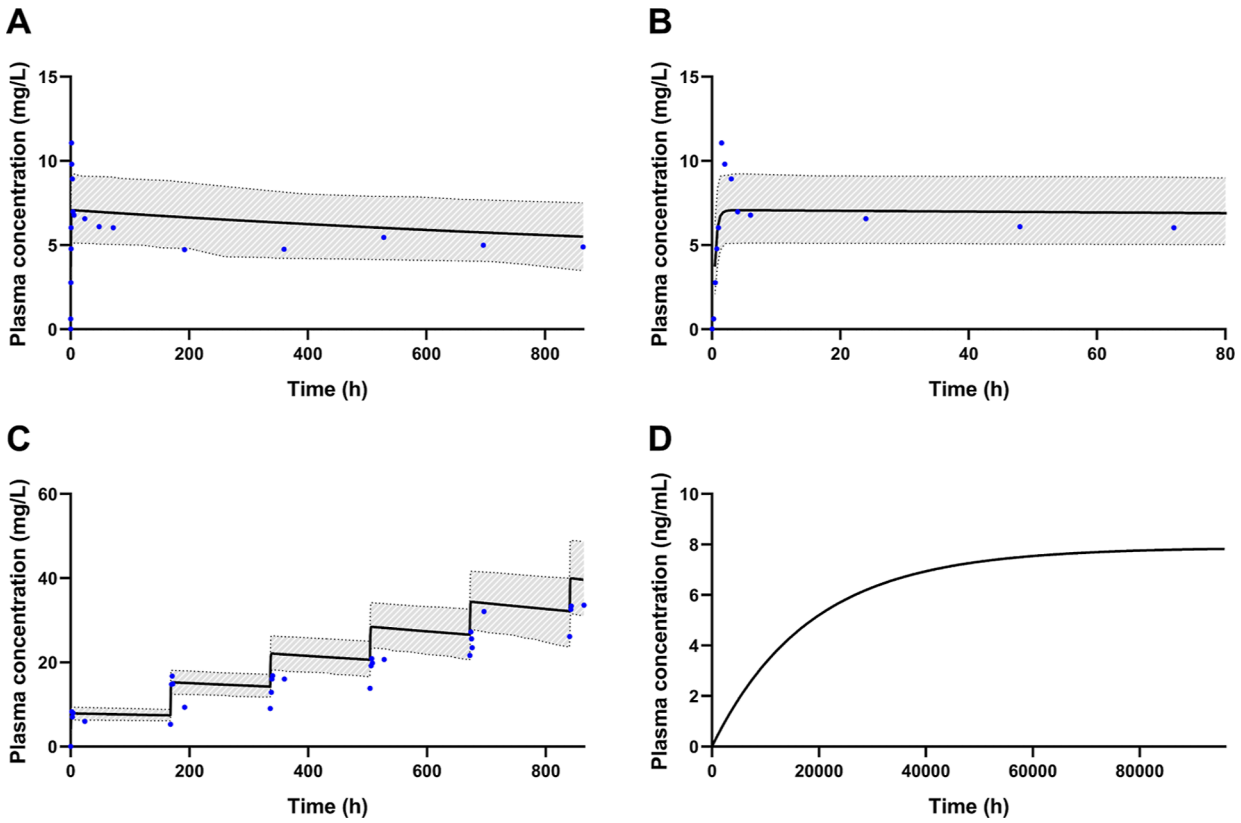

Figure 1. Simulated plasma concentration−time profile of PFOA at 50 mg single dose for (A) 864 and (B) 80 h, (C) 50 mg weekly dose for 864 h, and (D) average plasma concentration−time profile of a male and female population representative for 11 years-long of 70 ng daily dose. Black solid lines represent the predicted mean plasma concentrations of PFOA with the 95th and 5th percentiles bounded by grey shaded areas, and blue solid circles represent the clinical data from the phase I dose-escalation trial.

Table 1. Comparison between Clinical Data and Simulated TK Parameters of PFOA for Two Dosing Regimens

dosing regimen	50 mg single dose			50 mg weekly dose		
TK parameter[a]	clinical data	simulated data	predicted/observed ratio[b]	clinical data	simulated data	predicted/observed ratio[b]
AUC (mg/L·h)	4470.80	5268.57	**1.18**	4845.03	5518.36	**1.14**
C_{max} (mg/L)	9.25	6.97	**0.75**	27.24	34.07	**1.25**
T_{max} (h)	3.00	3.16	**1.05**	2.00	2.53	**1.27**
CL (L/h)	0.0112	0.0095	**0.85**	0.0103	0.0091	**0.88**
$t_{1/2}$ (h)	2773.90	2977.83	**1.07**			

[a]AUC, C_{max}, T_{max}, clearance (CL), and $t_{1/2}$ are expressed in geometric mean. [b]Predicted/Observed ratio refers to the ratio of simulated against clinical data.

Menten equation (eq 2) to account for simultaneous transport by two transporters.

To further isolate the kinetics of MCT1, inhibition studies were performed. As there is no MCT1-specific inhibitor that does not also inhibit other organic anion transporters, an indirect inhibition strategy using an OAT1-specific inhibitor was used. Figures S7B and S7C demonstrate that 2,4-dichlorophenoxyacetic acid (2,4-D) completely inhibited uptake of 4-aminohippuric acid (OAT1 probe substrate) at 200 μM in OAT1-transfected HEK293 cells, while Figure S7D shows identical PFOA uptake profiles for wild-type and OAT1-transfected cells upon inhibition of OAT1. This indicated the presence of a background transporter in OAT1-transfected cells and its functional capacity is comparable to that of the HEK293 wild-type. As each transport assay was always conducted using both wild-type and transfected HEK293 cells, for each experiment we obtained the background transport kinetics from the wild-type cells and substituted those values into eq 2 for isolation of PFOA uptake kinetics for the transporters of interest.

3.3. Relative Expression Factor Determination. Abundances of transporters stably expressed in HEK293 and CHO cell lines were quantified as described in Method S6 and are shown in Tables S5 and S6, respectively, along with previously reported in vivo transporter protein expression levels within human tissues (kidney cortex, cryopreserved isolated hepatocytes, and fresh human liver tissues). The calculated relative expression factor (REF) values for each transporter are determined as described in Method S7 and listed in Tables S5 and S6. The reported in vivo protein expression of hepatic transporters in both human liver and cryopreserved hepatocytes for the calculation of scaling factor 2 (SF2) is listed in Table S7.

3.4. Model Validation of PFOA and Application in Forward and Reverse Dosimetry. Simulated plasma concentration−time profiles and clinical data of PFOA at 50

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mg single and weekly doses in adult cancer population are shown in Figure 1A−C. The predicted and observed TK parameters as well as the predicted/observed ratios are tabulated in Table 1. A complete excel output of the Simcyp simulation results can be found in the Supporting Information. All predicted parameters fell within 2-fold of the observed clinical data for both single and weekly dosing regimens. The predicted $t_{1/2}$ (124 days) was slightly overestimated by 7% as compared to the observed clinical data for the 50 mg single dose.

A forward dosimetry simulation for daily exposure of 70 ng (representative of the average dietary intake for Canadian population) of PFOA over 11 years is illustrated in Figure 1D. Our predicted C_{ss} is 7.83 ng/mL, which overestimated the reported plasma concentration (2.52 ng/mL) in CHMS 2007−2009 with a predicted/observed ratio of 3.1. Our predicted $t_{1/2}$ is 2.0 years, within the range of the biomonitoring estimates (1.3−3.9 years). The predicted TK parameters of the male and female population representative and the resulting averages are tabulated in Table 2.

Table 2. Simulated TK Parameters of PFOA at 11 Years-Long of 70 ng Daily Dose for a Male and Female Population Representative and the Corresponding Averages

dosing regimen	70 ng daily dose for 11 years		
TK parameter	male population representative	female population representative	average values
C_{ss} (ng/mL)	8.37	7.28	7.83
$t_{1/2}$ (h)	18,574.34	15,768.55	17,171.45 (2.0 years)

Using our middle-out model for reverse dosimetry, the estimated daily intake that would result in the critical serum level of 2 ng/mL in an adult female is 0.28 ng/kg body weight as shown in Figure S8. Our estimate is higher than 0.187 ng/kg body weight as reported in EFSA Opinion 2020.

3.5. Sensitivity Analyses. Sensitivity analyses were performed as described in Method S8 to assess the effect of several drug-specific parameters, namely J_{max}/CL_{int} of hepatic and renal transporters as well as biliary clearance, on the TK of PFOA. The plasma concentration−time profiles are shown in Figure S9, while the values of drug parameters, as well as their respective clearance and AUC are tabulated in Table S8. PFOA clearance was heavily dependent on the renal apical uptake, basolateral uptake, and efflux transporters, and this dependency was also reflected in the drastic changes in plasma concentration observed as J_{max} of renal transporters was varied. On the other hand, changes to hepatic transporters and biliary clearance kinetics had minimal influence on the clearance and AUC. As for the intestinal absorption of PFOA, the result demonstrated significantly higher contribution from passive transport component (human jejunal effective permeability, $P_{eff,man}$) than active component (J_{max} of intestinal uptake transporter) as seen from the change in plasma concentration−time profile.

3.6. Model Applications. *3.6.1. Explaining the Differences in PFOA $t_{1/2}$ Observed in Clinical and Biomonitoring Studies.* There remains a large and unresolved difference in the reported $t_{1/2}$ between the clinical trial (representative of high-dose exposure) and biomonitoring studies (representative of low-dose exposure) which were 116 days[24] and 1.3−3.9 years,[7−13] respectively. Our model successfully recapitulated both observed $t_{1/2}$ values through the inclusion of a hypothetical renal basolateral efflux transporter that is saturable at high dose (identification is currently in progress). This change resulted in a material increase in the estimated $t_{1/2}$ of PFOA at low dose to 2.0 years, recapitulating the years-long estimates obtained from biomonitoring studies.

3.6.2. Assessing the Influence of Tubular Apical Uptake and Basolateral Efflux Processes in Mediating the Renal Reabsorption of PFOA. To investigate the role of active tubular reabsorption in influencing the clearance of PFOA, oral doses of PFOA were simulated in four scenarios: (1) in the presence of both renal apical uptake (OAT4 and URAT1) and basolateral efflux transporters, (2) basolateral efflux transporter only, (3) apical uptake transporters only, and (4) in the absence of both apical uptake and basolateral efflux transporters. The predicted TK parameters for the high dose with their respective ratios against clinical data and low-dose exposure are shown in Tables S9 and S10, respectively. For high-dose exposure, clearance was overpredicted, while $t_{1/2}$ and AUC were underestimated in scenarios (2) and (4). On the other hand, the predicted clearance, $t_{1/2}$, and AUC were all within 2-fold of the observed values for scenarios (1) and (3).

Since there was no clinical data available for comparison in the low-dose scenario, TK parameters predicted in scenario (1) was used to monitor the changes of TK parameters in the presence/absence of these transporters (Table S10). The simulations for scenarios (2) and (4) shared similar trends (overpredicted clearance and underestimated $t_{1/2}$) with their counterparts at high dose. Crucially, the removal of basolateral efflux transporter alone (scenario 3) resulted in a major decrease in the predicted $t_{1/2}$, in contrast to the unchanged $t_{1/2}$ when the same was performed for the high-dose regimen.

3.7. Investigating the Role of V_d in Determining the $t_{1/2}$ of PFOA. Apart from the liver and kidney, PFOA penetration into other tissues is modeled as perfusion-limited; however, our discovery of MCT1 suggests that other tissues also uptake PFOA in a permeability-limited manner. Without tissue-level MCT1 expression data, we were unable to construct permeability-limited tissue models. Nevertheless, to examine the potential impact of MCT1, we utilized the K_p scalar parameter which increases the tissue penetration of PFOA in a global manner, mimicking the effect of broad tissue uptake that might be mediated by MCT1 (see Method S9 for more details). As illustrated in Figure S10, $t_{1/2}$ remains mostly unchanged from K_p scalar of 0.5 to 2.3 and increases linearly against K_p scalar afterward, while V_d is linearly dependent on the K_p scalar. A K_p scalar of at least 3.541 (which implies 3.5× higher tissue penetration) is required to extend the $t_{1/2}$ to 2.3 years and above as reported in the biomonitoring studies. This also increases the predicted V_d to 0.214 L/kg, which is consistent with the reported V_d (0.170−0.191 L/kg).[21,47] These K_p scalar values were modified solely for model exploration purposes, and the default K_p scalar of 1 was applied in the final model structure and other simulations.

3.8. Importance of Quantitative Proteomics-Based IVIVE. We ran two simulations (with and without REF) to assess the importance of incorporating REF in the IVIVE of transporter kinetics, as this is not routinely employed in PBTK modeling. The estimated $t_{1/2}$ of PFOA from the biomonitoring studies and the predicted TK parameters with and without implementing REF are shown in Table S11. We observed a drastic decrease in our predicted clearance after incorporating

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REF into our simulations, resulting in an increase in the predicted $t_{1/2}$ from 0.16 to 2.12 years, which agreed with the observed data.

4. DISCUSSION

Here, we developed the first mechanistic, middle-out human PBTK model of PFOA to predict its TK profile and explain the rationale for its prolonged $t_{1/2}$ in humans. In particular, we extensively utilized in vitro kinetic data for liver and kidney transporters and performed robust IVIVE using quantitative proteomics. We successfully recapitulated the observed TK parameters and divergent $t_{1/2}$ from a clinical trial and biomonitoring studies and generated novel mechanistic insight into the disposition of PFOA.

Our model was validated with the recapitulation of the observed TK parameters (plasma concentrations, AUC, clearance, and estimated $t_{1/2}$) of the phase I dose-escalation trial at 50 mg single dose of PFOA. We were also able to recapitulate the plasma concentration profiles especially for the weekly dosing regimen (Figure 1C), indicating the robustness of our model. Since dietary intake is the major route of human exposure to PFOA,[3,48] we simulated the average dietary exposure for Canadians reported by Tittlemier et al.[42] Our predicted $t_{1/2}$ of 2.0 years recapitulated the estimates from biomonitoring studies. However, our predicted C_{ss} is higher than the reported plasma concentrations in CHMS 2007−2009, which indicates that our model is more conservative from a dosimetry perspective. We also performed reverse dosimetry to determine the daily intake of PFOA for an adult female that would result in C_{ss} of 2 ng/mL as reported by EFSA. Our predicted daily intake is higher than the reported estimate, indicating that the EFSA estimate is more conservative.

To date, the reasons behind the years-long $t_{1/2}$ of PFOA have remained unclear. From a pharmacokinetic perspective, an exaggerated $t_{1/2}$ is the result of either extensive tissue uptake and/or impaired clearance by the liver or kidney. Our model indicates that extensive transporter-mediated renal reabsorption is partly responsible for the long $t_{1/2}$ of PFOA observed in the biomonitoring studies. Sensitivity analyses of the kinetic parameters of hepatic and biliary transporters revealed minimal changes in the TK parameters as opposed to the dramatic changes seen for renal transporters. In particular, $t_{1/2}$ showed the largest dependence on the joint inclusion of apical uptake and basolateral efflux transporters in a low-dose scenario (Table S10) but was minimally changed for the hepatic transporters (data not shown). These suggest that renal transporters are a major contributor to the extraordinary retention of PFOA in humans.

We also identified that an increased volume of distribution, beyond that predicted by perfusion-limited tissue penetration, also contributed to the $t_{1/2}$ of PFOA. This is supported by our identification of MCT1 as a new transporter of PFOA. MCT1 is ubiquitously expressed in many human tissues such as the brain, heart, liver, kidney, muscles, skin, intestine, placenta, and testis.[49−51] This implies that the distribution of PFOA within the human body may be more widespread than is currently believed, as PFOA is ionized at physiological pH and requires transporters to mediate its tissue penetration, while the known transporters for PFOA are primarily expressed in the liver and kidney. It also explains observations from cadaver[52−54] and animal studies[55,56] that showed extensive tissue distribution of PFOA. By empirically increasing the K_p scalar parameter to mimic the effect of MCT1, we demonstrated that both the predicted $t_{1/2}$ and V_d closely matched the estimates from the biomonitoring studies. Interestingly, $t_{1/2}$ is relatively unchanged up to a K_p scalar of 2, which indicates that up to that point, clearance is the primary driver of $t_{1/2}$; beyond that, it is apparent that V_d begins dominating $t_{1/2}$. Importantly, if MCT1 is indeed mediating enhanced PFOA tissue penetration, then its V_d can be saturable in a high-dose scenario and, in a manner similar to $t_{1/2}$, explains the different estimates of V_d in the clinical study by Elcombe et al. (0.08 L/kg, determined by the authors as described in Method 2.4) and the biomonitoring studies (0.17−0.19 L/kg). To refine the prediction of tissue PFOA levels and $t_{1/2}$, it will be important to incorporate MCT1 transport and expression into our model (ongoing work). The susceptibility of other PFAS to MCT1 transport also needs further investigation.

A key unresolved controversy relates to the large difference between the $t_{1/2}$ estimated from the biomonitoring studies and the PFOA clinical trial.[24] All currently known renal transporters (OAT1, OAT3, OAT4, and URAT1) function to pump PFOA into proximal tubule cells (PTCs), but being highly charged at physiological pH, only a small amount of PFOA can passively exit the PTCs. We explored a mechanism to remove PFOA from PTCs to prevent excessive buildup of PFOA using the construct of a hypothetical basolateral efflux transporter. We further allowed this transporter to be saturated at high intra-PTC concentrations (which occurs in a high-dose scenario), resulting in its inhibition. This in turn markedly limits renal reabsorption, hence shortening the $t_{1/2}$ of PFOA. Conversely, at a low-dose exposure, the transporter is not saturated, permitting enhanced retention of PFOA and resulting in a longer $t_{1/2}$. This proposed mechanism is summarized in Figure S11. Our simulations predicted the $t_{1/2}$ of PFOA to be 124 days at high dose and 2.0 years at low dose, providing a plausible mechanism to unify findings from both the biomonitoring studies and clinical trial. Work is currently underway to identify and characterize this transporter in relation to PFOA efflux.

Similar to the kidney, it was necessary to include a hypothetical sinusoidal efflux transporter to permit re-entry of PFOA from liver into the blood and avoid unrealistic, perpetual PFOA accumulation in the liver. The absence of this hypothetical sinusoidal efflux transporter results in a major decrease in plasma PFOA levels as massive amounts of PFOA are trapped within the liver which cannot be eliminated (Figure S12). While canalicular removal of PFOA through the biliary route is possible,[57] the estimated contribution of this pathway is very low and sensitivity analyses of our model indicate that it does not modify the plasma concentrations of PFOA (Figure S9F). In addition, as PFOA that undergoes biliary excretion is completely reabsorbed, it does not contribute to systemic clearance.

While PBTK modeling is reasonably accurate for metabolism-dependent chemicals, its use for transport-dependent compounds is less established. Our model demonstrates a mechanistic approach to simultaneously account for the effects of 10 transporters on chemical disposition. To our knowledge, this is the most extensive effort in TK modeling involving transporters. Quantitative proteomics and application of REF allowed the correction of significantly lower absolute transporter expression levels observed within our cultured mammalian cells compared with human tissues (Tables S5 and S6), which would otherwise affect the accuracy of the

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kinetic data used for modeling. These results demonstrate that applying quantitative proteomics-based IVIVE is critical to improve the accuracy of PBTK models for transport-dependent compounds.

To the best of our knowledge, the $t_{1/2}$ estimates of PFOA used for risk assessment mostly originated from biomonitoring studies. However, there are limitations with their $t_{1/2}$ determination. For example, in deriving the estimated $t_{1/2}$ for PFOA, Zhang et al.[10] made several assumptions, such as the use of a single urine concentration instead of a 24 h urine collection and a standard V_d value for all PFAS studied, which may not be accurate and introduces uncertainties into the estimates. Here, we demonstrated that our PBTK model can overcome these limitations. Our PBTK model can be considered as an independent methodology that is complementary to biomonitoring estimates and provides deep mechanistic insight into the TK for PFOA.

Despite key advances, there remain limitations in our model. First, we were not able to fully recapitulate the observed plasma concentration−time profile during the first few hours post-dosing for the 50 mg single dose as shown in Figure 1B. However, we believe that the observed time course during the first few hours post-dosing needs to consider the small sample size (three patients were studied for this dosing regimen), which may not capture the population-level profile. In addition, it is possible that at such high plasma concentrations, MCT1-mediated tissue uptake is saturated, which would result in augmented plasma concentrations. Hence, the deviation in our simulation from the observed data does not diminish the accuracy of our simulations. Second, extensive validation of various TK parameters of our model was not possible due to limited available clinical data and the relatively small sample size for each dosing regimen used in the phase I dose-escalation trial. Third, interpretation on the TK estimates for PFOA from biomonitoring studies discussed in this article should be taken with caution due to the level of uncertainty of ongoing exposure inherently present in such studies. Finally, although MCT1 is ubiquitously expressed in many organs, the lack of tissue-level quantitative expressions precludes its inclusion as an uptake transporter in these organs within our model. As such data becomes available, the model can be refined to assess the influence of MCT1 in driving tissue penetration of PFOA and more accurately estimate its V_d and $t_{1/2}$.

In conclusion, our study presents the first systematic PBTK model that comprehensively integrates kinetic data from all known PFOA transporters with the aid of quantitative proteomics-based IVIVE to explain the prolonged biological $t_{1/2}$ of PFOA. Importantly, close attention was paid to the input data quality, and where necessary, fresh kinetic data were generated to eliminate the confounding effects of MCT1. We also provided the first mechanistic explanation of the divergent PFOA $t_{1/2}$ measured from the clinical trial and biomonitoring studies, and our model highlighted the importance of renal reabsorption in controlling its elimination. Notably, our work has led to the discovery of MCT1 as a potentially new PFOA transporter, with potential implications for enhanced tissue distribution and reverse dosimetry determinations, as well as a yet uncharacterized renal basolateral efflux transporter. Finally, having constructed and validated a generic PBTK model for a data-rich PFAS such as PFOA, we anticipate that our approach can be extended to predict human TK profiles for other data-poor PFAS through the same IVIVE workflow. Work is underway in our laboratory to expand our model to other PFAS compounds.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.2c05642.

Materials and methods for cell culture, in vitro uptake assays of PFOA and model simulations, results for REF determination, sensitivity analyses and assessment of importance of quantitative proteomics-based IVIVE, additional figures and tables presenting data for uptake assays and model simulation outcomes, compound parameters for the model and LC−MS/MS parameters (PDF)

Simcyp compound file for PFOA (ZIP)

Simulation output for 50 mg single dose (ZIP)

Simulation output for male representative with 70 ng daily dose (ZIP)

Simulation output for female representative with 70 ng daily dose (ZIP)

Simulation output for female representative with daily dose of 0.28 ng per kg body weight (ZIP)

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J.L, S.Y.C, and J.C.Y.C wrote the article. J.L, S.Y.C, S.P.F.T, H.C.K, E.J.Y.C, E.C.Y.C, and J.C.Y.C designed the research. J.L, S.Y.C, S.P.F.T, H.C.K, and E.J.Y.C conducted the experiments. J.L, S.Y.C, S.P.F.T, H.C.K, and J.C.Y.C performed the data analysis.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Buck, R. C.; Franklin, J.; Berger, U.; Conder, J. M.; Cousins, I. T.; de Voogt, P.; Jensen, A. A.; Kannan, K.; Mabury, S. A.; van Leeuwen, S. P. Perfluoroalkyl and Polyfluoroalkyl Substances in the Environment: Terminology, Classification, and Origins. *Integr. Environ. Assess. Manage.* **2011**, *7*, 513−541.

(2) Vestergren, R.; Cousins, I. T.; Trudel, D.; Wormuth, M.; Scheringer, M. Estimating the Contribution of Precursor Compounds in Consumer Exposure to PFOS and PFOA. *Chemosphere* **2008**, *73*, 1617−1624.

(3) Sunderland, E. M.; Hu, X. C.; Dassuncao, C.; Tokranov, A. K.; Wagner, C. C.; Allen, J. G. A Review of the Pathways of Human Exposure to Poly- and Perfluoroalkyl Substances (PFASs) and Present Understanding of Health Effects. *J. Exposure Sci. Environ. Epidemiol.* **2019**, *29*, 131−147.

(4) Calafat, A. M.; Wong, L. Y.; Kuklenyik, Z.; Reidy, J. A.; Needham, L. L. Polyfluoroalkyl Chemicals in the U.S. Population: Data from the National Health and Nutrition Examination Survey (NHANES) 2003-2004 and Comparisons with NHANES 1999-2000. *Environ. Health Perspect.* **2007**, *115*, 1596−1602.

(5) Fromme, H.; Tittlemier, S. A.; Völkel, W.; Wilhelm, M.; Twardella, D. Perfluorinated Compounds − Exposure Assessment for the General Population in Western Countries. *Int. J. Hyg. Environ. Health* **2009**, *212*, 239−270.

(6) Monroy, R.; Morrison, K.; Teo, K.; Atkinson, S.; Kubwabo, C.; Stewart, B.; Foster, W. G. Serum Levels of Perfluoroalkyl Compounds in Human Maternal and Umbilical Cord Blood Samples. *Environ. Res.* **2008**, *108*, 56−62.

(7) Olsen, G. W.; Burris, J. M.; Ehresman, D. J.; Froehlich, J. W.; Seacat, A. M.; Butenhoff, J. L.; Zobel, L. R. Half-Life of Serum Elimination of Perfluorooctanesulfonate, Perfluorohexanesulfonate, and Perfluorooctanoate in Retired Fluorochemical Production Workers. *Environ. Health Perspect.* **2007**, *115*, 1298−1305.

(8) Brede, E.; Wilhelm, M.; Göen, T.; Müller, J.; Rauchfuss, K.; Kraft, M.; Hölzer, J. Two-Year Follow-up Biomonitoring Pilot Study of Residents' and Controls' PFC Plasma Levels after PFOA Reduction in Public Water System in Arnsberg, Germany. *Int. J. Hyg. Environ. Health* **2010**, *213*, 217−223.

(9) Bartell, S. M.; Calafat, A. M.; Lyu, C.; Kato, K.; Ryan, P. B.; Steenland, K. Rate of Decline in Serum PFOA Concentrations after Granular Activated Carbon Filtration at Two Public Water Systems in Ohio and West Virginia. *Environ. Health Perspect.* **2010**, *118*, 222−228.

(10) Zhang, Y.; Beesoon, S.; Zhu, L.; Martin, J. W. Biomonitoring of Perfluoroalkyl Acids in Human Urine and Estimates of Biological Half-Life. *Environ. Sci. Technol.* **2013**, *47*, 10619−10627.

(11) Gomis, M. I.; Vestergren, R.; Nilsson, H.; Cousins, I. T. Contribution of Direct and Indirect Exposure to Human Serum Concentrations of Perfluorooctanoic Acid in an Occupationally Exposed Group of Ski Waxers. *Environ. Sci. Technol.* **2016**, *50*, 7037−7046.

(12) Worley, R. R.; Moore, S. M. A.; Tierney, B. C.; Ye, X.; Calafat, A. M.; Campbell, S.; Woudneh, M. B.; Fisher, J. Per- and Polyfluoroalkyl Substances in Human Serum and Urine Samples from a Residentially Exposed Community. *Environ. Int.* **2017**, *106*, 135−143.

(13) Li, Y.; Fletcher, T.; Mucs, D.; Scott, K.; Lindh, C. H.; Tallving, P.; Jakobsson, K. Half-Lives of PFOS, PFHxS and PFOA after End of Exposure to Contaminated Drinking Water. *Occup. Environ. Med.* **2018**, *75*, 46−51.

(14) Lau, C.; Anitole, K.; Hodes, C.; Lai, D.; Pfahles-Hutchens, A.; Seed, J. Perfluoroalkyl Acids: A Review of Monitoring and Toxicological Findings. *Toxicol. Sci.* **2007**, *99*, 366−394.

(15) Steenland, K.; Fletcher, T.; Savitz, D. A. Epidemiologic Evidence on the Health Effects of Perfluorooctanoic Acid (PFOA). *Environ. Health Perspect.* **2010**, *118*, 1100−1108.

(16) Kataria, A.; Trachtman, H.; Malaga-Dieguez, L.; Trasande, L. Association between Perfluoroalkyl Acids and Kidney Function in a Cross-Sectional Study of Adolescents. *Environ. Health* **2015**, *14*, 89.

(17) Burns, D. C.; Ellis, D. A.; Li, H.; McMurdo, C. J.; Webster, E. Experimental PKa Determination for Perfluorooctanoic Acid (PFOA) and the Potential Impact of PKa Concentration Dependence on Laboratory-Measured Partitioning Phenomena and Environmental Modeling. *Environ. Sci. Technol.* **2008**, *42*, 9283−9288.

(18) Wetmore, B. A.; Wambaugh, J. F.; Ferguson, S. S.; Sochaski, M. A.; Rotroff, D. M.; Freeman, K.; Clewell, H. J.; Dix, D. J.; Andersen, M. E.; Houck, K. A.; Allen, B.; Judson, R. S.; Singh, R.; Kavlock, R. J.; Richard, A. M.; Thomas, R. S. Integration of Dosimetry, Exposure, and High-Throughput Screening Data in Chemical Toxicity Assessment. *Toxicol. Sci.* **2012**, *125*, 157−174.

(19) Zhao, W.; Zitzow, J. D.; Weaver, Y.; Ehresman, D. J.; Chang, S. C.; Butenhoff, J. L.; Hagenbuch, B. Organic Anion Transporting

Polypeptides Contribute to the Disposition of Perfluoroalkyl Acids in Humans and Rats. *Toxicol. Sci.* **2017**, *156*, 84−95.
(20) Ruggiero, M. J.; Miller, H.; Idowu, J. Y.; Zitzow, J. D.; Chang, S.-C.; Hagenbuch, B. Perfluoroalkyl Carboxylic Acids Interact with the Human Bile Acid Transporter NTCP. *Livers* **2021**, *1*, 221−229.
(21) Han, X.; Nabb, D. L.; Russell, M. H.; Kennedy, G. L.; Rickard, R. W. Renal Elimination of Perfluorocarboxylates (PFCAs). *Chem. Res. Toxicol.* **2012**, *25*, 35−46.
(22) Worley, R. R.; Fisher, J. Application of Physiologically-Based Pharmacokinetic Modeling to Explore the Role of Kidney Transporters in Renal Reabsorption of Perfluorooctanoic Acid in the Rat. *Toxicol. Appl. Pharmacol.* **2015**, *289*, 428−441.
(23) Campbell, J.; Clewell, H.; Cox, T.; Dourson, M.; Ethridge, S.; Forsberg, N.; Gadagbui, B.; Hamade, A.; Naidu, R.; Pechacek, N.; Peixe, T. S.; Prueitt, R.; Rachamalla, M.; Rhomberg, L.; Smith, J.; Verma, N. The Conundrum of the PFOA Human Half-Life, an International Collaboration. *Regul. Toxicol. Pharmacol.* **2022**, *132*, 105185.
(24) Elcombe, C. R.; Wolf, C. R.; Westwood, A. L. Compositions Comprising Perfluorooctanoic Acid. International Patent. WO 2011101643 A1, 2011.
(25) Loccisano, A. E.; Campbell, J. L.; Andersen, M. E.; Clewell, H. J. Evaluation and Prediction of Pharmacokinetics of PFOA and PFOS in the Monkey and Human Using a PBPK Model. *Regul. Toxicol. Pharmacol.* **2011**, *59*, 157−175.
(26) Loccisano, A. E.; Longnecker, M. P.; Campbell, J. L.; Andersen, M. E.; Clewell, H. J. Development of Pbpk Models for Pfoa and Pfos for Human Pregnancy and Lactation Life Stages. *J. Toxicol. Environ. Health, Part A* **2013**, *76*, 25−57.
(27) Worley, R. R.; Yang, X.; Fisher, J. Physiologically Based Pharmacokinetic Modeling of Human Exposure to Perfluorooctanoic Acid Suggests Historical Non Drinking-Water Exposures Are Important for Predicting Current Serum Concentrations. *Toxicol. Appl. Pharmacol.* **2017**, *330*, 9−21.
(28) Schrenk, D.; Bignami, M.; Bodin, L.; Chipman, J. K.; del Mazo, J.; Grasl-Kraupp, B.; Hogstrand, C.; Hoogenboom, L.; Leblanc, J.-C.; Nebbia, C. S.; Nielsen, E.; Ntzani, E.; Petersen, A.; Sand, S.; Vleminckx, C.; Wallace, H.; Barregård, L.; Ceccatelli, S.; Cravedi, J.; Halldorsson, T. I.; Haug, L. S.; Johansson, N.; Knutsen, H. K.; Rose, M.; Roudot, A.; Van Loveren, H.; Vollmer, G.; Mackay, K.; Riolo, F.; Schwerdtle, T. Risk to Human Health Related to the Presence of Perfluoroalkyl Substances in Food. *EFSA J.* **2020**, *18*, 6223.
(29) Schneckener, S.; Preuss, T. G.; Kuepfer, L.; Witt, J. A Workflow to Build PBTK Models for Novel Species. *Arch. Toxicol.* **2020**, *94*, 3847−3860.
(30) Chan, J. C. Y.; Tan, S. P. F.; Upton, Z.; Chan, E. C. Y. Bottom-up Physiologically-Based Biokinetic Modelling as an Alternative to Animal Testing. *ALTEX* **2019**, *36*, 597−612.
(31) Jamei, M.; Marciniak, S.; Feng, K.; Barnett, A.; Tucker, G.; Rostami-Hodjegan, A. The Simcyp Population-Based ADME Simulator. *Expert Opin. Drug Metab. Toxicol.* **2009**, *5*, 211−223.
(32) Rowland Yeo, K.; Jamei, M.; Yang, J.; Tucker, G. T.; Rostami-Hodjegan, A. Physiologically Based Mechanistic Modelling to Predict Complex Drug−Drug Interactions Involving Simultaneous Competitive and Time-Dependent Enzyme Inhibition by Parent Compound and Its Metabolite in Both Liver and Gut—The Effect of Diltiazem on the Time-Course of Exposure to Triazolam. *Eur. J. Pharm. Sci.* **2010**, *39*, 298−309.
(33) Ezuruike, U.; Zhang, M.; Pansari, A.; De Sousa Mendes, M.; Pan, X.; Neuhoff, S.; Gardner, I. Guide to Development of Compound Files for PBPK Modeling in the Simcyp Population-based Simulator. *CPT: Pharmacometrics Syst. Pharmacol.* **2022**, *11*, 805−821.
(34) Jamei, M.; Turner, D.; Yang, J.; Neuhoff, S.; Polak, S.; Rostami-Hodjegan, A.; Tucker, G. Population-Based Mechanistic Prediction of Oral Drug Absorption. *AAPS J.* **2009**, *11*, 225−237.
(35) Jamei, M.; Bajot, F.; Neuhoff, S.; Barter, Z.; Yang, J.; Rostami-Hodjegan, A.; Rowland-Yeo, K. A Mechanistic Framework for In Vitro−In Vivo Extrapolation of Liver Membrane Transporters: Prediction of Drug−Drug Interaction Between Rosuvastatin and Cyclosporine. *Clin. Pharmacokinet.* **2014**, *53*, 73−87.
(36) Burt, H. J.; Neuhoff, S.; Almond, L.; Gaohua, L.; Harwood, M. D.; Jamei, M.; Rostami-Hodjegan, A.; Tucker, G. T.; Rowland-Yeo, K. Metformin and Cimetidine: Physiologically Based Pharmacokinetic Modelling to Investigate Transporter Mediated Drug−Drug Interactions. *Eur. J. Pharm. Sci.* **2016**, *88*, 70−82.
(37) Scotcher, D.; Jones, C. R.; Galetin, A.; Rostami-Hodjegan, A. Delineating the Role of Various Factors in Renal Disposition of Digoxin through Application of Physiologically Based Kidney Model to Renal Impairment Populations. *J. Pharmacol. Exp. Ther.* **2017**, *360*, 484−495.
(38) Rodgers, T.; Leahy, D.; Rowland, M. Physiologically Based Pharmacokinetic Modeling 1: Predicting the Tissue Distribution of Moderate-to-Strong Bases. *J. Pharm. Sci.* **2005**, *94*, 1259−1276.
(39) Rodgers, T.; Rowland, M. Physiologically-Based Pharmacokinetic Modeling 2: Predicting the Tissue Distribution of Acids, Very Weak Bases, Neutrals and Zwitterions. *J. Pharm. Sci.* **2007**, *95*, 1238.
(40) Rodgers, T.; Rowland, M. Mechanistic Approaches to Volume of Distribution Predictions: Understanding the Processes. *Pharm. Res.* **2007**, *24*, 918−933.
(41) Chen, Y.; Jin, J. Y.; Mukadam, S.; Malhi, V.; Kenny, J. R. Application of IVIVE and PBPK Modeling in Prospective Prediction of Clinical Pharmacokinetics: Strategy and Approach during the Drug Discovery Phase with Four Case Studies. *Biopharm. Drug Dispos.* **2012**, *33*, 85−98.
(42) Tittlemier, S. A.; Pepper, K.; Seymour, C.; Moisey, J.; Bronson, R.; Cao, X. L.; Dabeka, R. W. Dietary Exposure of Canadians to Perfluorinated Carboxylates and Perfluorooctane Sulfonate via Consumption of Meat, Fish, Fast Foods, and Food Items Prepared in Their Packaging. *J. Agric. Food Chem.* **2007**, *55*, 3203−3210.
(43) Health Canada. Report on Human Biomonitoring of Environmental Chemicals in Canada: Results of the Canadian Health Measures Survey Cycle 1 (2007−2009), 2010; Available from. http://www.hc-sc.gc.ca/ewh-semt/pubs/contaminants/chms-ecms/index-eng.php (accessed Nov 12, 2022).
(44) Ahlin, G.; Hilgendorf, C.; Karlsson, J.; Szigyarto, C. A. K.; Uhlén, M.; Artursson, P. Endogenous Gene and Protein Expression of Drug-Transporting Proteins in Cell Lines Routinely Used in Drug Discovery Programs. *Drug Metab. Dispos.* **2009**, *37*, 2275−2283.
(45) Garcia, C. K.; Goldstein, J. L.; Pathak, R. K.; Anderson, R. G. W.; Brown, M. S. Molecular Characterization of a Membrane Transporter for Lactate, Pyruvate, and Other Monocarboxylates: Implications for the Cori Cycle. *Cell* **1994**, *76*, 865−873.
(46) Halestrap, A. P. Monocarboxylic Acid Transport. *Compr. Physiol.* **2013**, *3*, 1611−1643.
(47) Thompson, J.; Lorber, M.; Toms, L. M. L.; Kato, K.; Calafat, A. M.; Mueller, J. F. Use of Simple Pharmacokinetic Modeling to Characterize Exposure of Australians to Perfluorooctanoic Acid and Perfluorooctane Sulfonic Acid. *Environ. Int.* **2010**, *36*, 390−397.
(48) Poothong, S.; Papadopoulou, E.; Padilla-Sánchez, J. A.; Thomsen, C.; Haug, L. S. Multiple Pathways of Human Exposure to Poly- and Perfluoroalkyl Substances (PFASs): From External Exposure to Human Blood. *Environ. Int.* **2020**, *134*, 105244.
(49) Halestrap, A. P.; Price, N. T. The Proton-Linked Monocarboxylate Transporter (MCT) Family: Structure, Function and Regulation. *Biochem. J.* **1999**, *343*, 281−299.
(50) Bonen, A.; Heynen, M.; Hatta, H. Distribution of Monocarboxylate Transporters MCT1-MCT8 in Rat Tissues and Human Skeletal Muscle. *Appl. Physiol., Nutr., Metab.* **2006**, *31*, 31−39.
(51) Fisel, P.; Schaeffeler, E.; Schwab, M. Clinical and Functional Relevance of the Monocarboxylate Transporter Family in Disease Pathophysiology and Drug Therapy. *Clin. Transl. Sci.* **2018**, *11*, 352−364.
(52) Maestri, L.; Negri, S.; Ferrari, M.; Ghittori, S.; Fabris, F.; Danesino, P.; Imbriani, M. Determination of Perfluorooctanoic Acid and Perfluorooctanesulfonate in Human Tissues by Liquid Chromatography/Single Quadrupole Mass Spectrometry. *Rapid Commun. Mass Spectrom.* **2006**, *20*, 2728−2734.

(53) Pérez, F.; Nadal, M.; Navarro-Ortega, A.; Fàbrega, F.; Domingo, J. L.; Barceló, D.; Farré, M. Accumulation of Perfluoroalkyl Substances in Human Tissues. *Environ. Int.* **2013**, *59*, 354−362.

(54) Koskela, A.; Koponen, J.; Lehenkari, P.; Viluksela, M.; Korkalainen, M.; Tuukkanen, J. Perfluoroalkyl Substances in Human Bone: Concentrations in Bones and Effects on Bone Cell Differentiation. *Sci. Rep.* **2017**, *7*, 6841.

(55) Bogdanska, J.; Borg, D.; Bergström, U.; Mellring, M.; Bergman, Å.; DePierre, J.; Nobel, S. Tissue Distribution of 14C-Labelled Perfluorooctanoic Acid in Adult Mice after 1−5 Days of Dietary Exposure to an Experimental Dose or a Lower Dose That Resulted in Blood Levels Similar to Those Detected in Exposed Humans. *Chemosphere* **2020**, *239*, 124755.

(56) Bian, Y.; He, M.-Y.; Ling, Y.; Wang, X.-J.; Zhang, F.; Feng, X.-S.; Zhang, Y.; Xing, S.-G.; Li, J.; Qiu, X.; Li, Y.-R. Tissue Distribution Study of Perfluorooctanoic Acid in Exposed Zebrafish Using MALDI Mass Spectrometry Imaging. *Environ. Pollut.* **2022**, *293*, 118505.

(57) Harada, K. H.; Hashida, S.; Kaneko, T.; Takenaka, K.; Minata, M.; Inoue, K.; Saito, N.; Koizumi, A. Biliary Excretion and Cerebrospinal Fluid Partition of Perfluorooctanoate and Perfluorooctane Sulfonate in Humans. *Environ. Toxicol. Pharmacol.* **2007**, *24*, 134−139.

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